

Algebra II with Trigonometry

Chapter 5.6

Olympiad Solutions (Gemini)

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Solutions

1. In $\triangle ABC$, the measure of $\angle C$ is 60° . The side lengths a and b differ by exactly 3, and side c has a length of 7. Determine the lengths of sides a and b .

By the Law of Cosines, we have: $c^2 = a^2 + b^2 - 2ab \cos C$

Substitute the given values $c = 7$ and $\angle C = 60^\circ$ (so $\cos 60^\circ = \frac{1}{2}$):

$$49 = a^2 + b^2 - ab$$

We can cleverly rewrite the right side in terms of the difference of the sides:

$$49 = (a - b)^2 + ab$$

We are given that $|a - b| = 3$, so $(a - b)^2 = 9$. Substituting this into the equation yields: $49 = 9 + ab \implies ab = 40$

We now seek two positive numbers whose absolute difference is 3 and whose product is 40. These numbers are the roots of the quadratic equation $t^2 - 3t - 40 = 0$, which factors as $(t - 8)(t + 5) = 0$. Discarding the negative root, we find the lengths of the two sides to be exactly 8 and 5.

2. A triangle has side lengths $13x$, $14x$, and $15x$ for some positive real number x . If its area is numerically equal to its semiperimeter, determine the exact value of x .

For a reference triangle with side lengths 13, 14, and 15, the semiperimeter is: $s = \frac{13+14+15}{2} = 21$

By Heron's formula, the area of this reference triangle is:

$$A = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{21(8)(7)(6)} = \sqrt{7056} = 84$$

When the sides are scaled by a factor of x , the semiperimeter scales linearly and the area scales quadratically. Thus, the new semiperimeter is $21x$ and the new area is $84x^2$.

Setting the area equal to the semiperimeter gives: $84x^2 = 21x$

Since x is strictly positive, we divide by $21x$ to obtain: $4x = 1 \implies x = \frac{1}{4}$

3. In quadrilateral $ABCD$, the side lengths are $AB = 7$, $BC = 24$, $CD = 20$, and $DA = 15$. Given that $\angle ABC$ is exactly 90° , determine the measure of $\angle ADC$ without writing down any calculations.

Observe the right triangle $\triangle ABC$ formed by the diagonal AC . The legs are $AB = 7$ and $BC = 24$. Recognizing the well-known 7-24-25 Pythagorean triple, the hypotenuse AC must be 25.

Now consider $\triangle ADC$, bounded by sides $DA = 15$, $CD = 20$, and the newly found diagonal $AC = 25$. These side lengths fall perfectly into the ratio $15 : 20 : 25$, which simplifies to $3 : 4 : 5$. Because this is a Pythagorean triple, $\triangle ADC$ is a right triangle.

The longest side is $AC = 25$, meaning the angle directly opposite it must be the right angle. Therefore, the measure of $\angle ADC$ is 90° .

4. A ship leaves port and sails 40 miles at a bearing of 020° (measured clockwise from due North). It then turns and sails another 40 miles at a bearing of 140° . What is the ship's exact straight-line distance from the port?

Let the port be point P , the turning point be T , and the final destination be F . The ship travels along segments PT and TF , each of length 40.

The ship changes its direction from a bearing of 020° to 140° . The change in bearing establishes the exterior angle of the ship's turn at T , which is $140^\circ - 20^\circ = 120^\circ$. Consequently, the interior angle of the triangle at T is the supplement: $180^\circ - 120^\circ = 60^\circ$.

We now have $\triangle PTF$ with two equal sides ($PT = TF = 40$) and an included angle of 60° . An isosceles triangle with a 60° vertex angle is, by necessity, an equilateral triangle. Therefore, the third side PF is equal to the other two, making the ship's straight-line distance from the port exactly 40 miles.

5. The side lengths a , b , and c of a triangle satisfy the algebraic relation $2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4) = 576$. Mentally determine the area of the triangle.

The algebraic expression given on the left side of the equation is precisely the radicand in the expanded form of Heron's formula. Specifically, the area A of a triangle can be expressed as:

$$A = \frac{1}{4} \sqrt{2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4)}$$

Squaring both sides and multiplying by 16 provides the identity:

$$16A^2 = 2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4)$$

Given that the right side evaluates to 576, we have: $16A^2 = 576$

Dividing by 16 mentally gives $A^2 = 36$. Since the area is positive, the area of the triangle is 6.

6. The side lengths of an oblique triangle are given by the expressions $2x + 1$, $x^2 - 1$, and $x^2 + x + 1$ for some integer $x > 1$. Determine the degree measure of the largest angle in this triangle.

Let $a = 2x + 1$, $b = x^2 - 1$, and $c = x^2 + x + 1$. To identify the longest side, notice that: $c - a = x^2 - x = x(x - 1) > 0$ (since $x > 1$)
 $c - b = x + 2 > 0$ Thus, c is strictly the longest side, and the largest angle C is opposite side c .

We apply the Law of Cosines to find $\angle C$: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

Instead of a brute-force expansion, we can elegantly calculate the numerator using the difference of squares:

$$b^2 - c^2 = (b - c)(b + c) = (-x - 2)(2x^2 + x) = -x(x + 2)(2x + 1)$$

Now, add $a^2 = (2x + 1)^2$ to this result:

$$a^2 + b^2 - c^2 = (2x + 1)^2 - x(x + 2)(2x + 1) \text{ Factor out } (2x + 1):$$

$$a^2 + b^2 - c^2 = (2x + 1)[(2x + 1) - (x^2 + 2x)] = (2x + 1)(1 - x^2)$$

Notice that $a = 2x + 1$ and $-b = 1 - x^2$. Thus, the numerator perfectly simplifies to $-ab$.

Substitute this back into the Law of Cosines: $\cos C = \frac{-ab}{2ab} = -\frac{1}{2}$

The angle with a cosine of $-\frac{1}{2}$ in a triangle is exactly 120° .